

St. Francis Institute of Technology
Department of information technology

Academic year 2024-25

Class SEIT A/B SEM: IV EXPERIMENT 2 : Python Constructs

1. AIM :

a) To implement a Python program to using Conditional statements (demo of even/odd, positive/negative/zero number, demo of greater between two/three numbers, divisibility)

b) To implement a Python program using Loop statements (demo of print 10 numbers, summation of 10 random numbers, range function , multiplication table etc.)

2. Objectives: After performing this experiment, a student will be able to write a basic Python program using various python constructs

3. Prerequisite: basics of python programming, algorithms

4. Requirement: PC, Python 3.9, Windows 10/ MacOS/ Linux , IDE

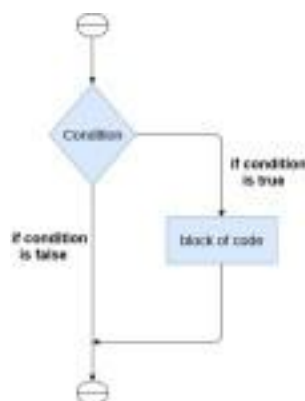
5. Pre-Experiment Exercise:

Theory :

a) Conditional statements/constructs

If Statement : The if statement is used to test a specific condition. If the condition is true, a block of code (if-block) will be executed.

If - else Statement: The if-else statement is similar to if statement except the fact that, it also provides the block of the code for the false case of the condition to be checked. If the condition provided in the if statement is false, then the else statement will be executed. **Nested if Statement:** Nested if statements enable us to use if - else statement inside an outer if statement.



The syntax of the if-statement is given below.

if expression:

statement #the block of statements after if statement will always follow indentation

eg: (note condition checking is done using relational operators, multiple conditions can be combined with logical operators)

```
if num%2 == 0:  
    print("The Given number is an even number")
```

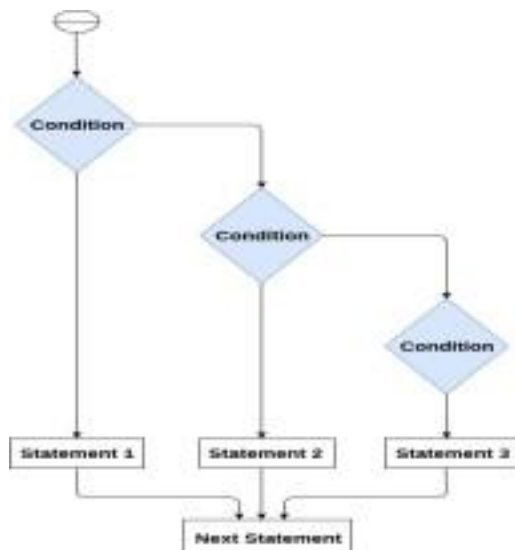
```
if a>b and a>c:  
    print("a is greater")  
else:  
    print("a is not greater")
```

The elif statement

The elif statement enables us to check multiple conditions and execute the specific block of statements depending upon the true condition among them.

Eg (note elif should be exactly aligned to if for indentation else error will be shown) if expression 1:

```
    # block of statements  
elif expression 2:  
    # block of statements  
elif expression 3:  
    # block of statements  
else:  
    # block of statements
```



b) Loop statements – for, while

For loop :

This type of loop executes a code block multiple times and abbreviates the code that manages the loop variable.

Syntax of the for Loop

for value **in** sequence:

code block

eg :

```
numbers = [4, 2, 6, 7, 3, 5, 8, 10, 6, 1, 9, 2]
```

for value **in** numbers:

```
square = value ** 2
```

```
print("The list of { } is { }", value, squares)
```

While loop :

Repeats a statement or group of statements while a given condition is TRUE. It tests the condition before executing the loop body

Syntax of the while loop is:

```
while <condition>:
```

```
{ code block }
```

Eg:

```
# Python program to show how to use a while loop
```

```
counter = 0
```

```
# Initiating the loop
```

```
while counter < 10: # giving the condition
```

```
    counter = counter + 3
```

```
    print("Python Loops")
```

(note we can use for and while with else statement too)

Loop Control Statements

- **Continue** : It returns the control to the beginning of the loop.
- **Break**: It stops the execution of the loop when the break statement is reached.
- **Pass statements** are used to create empty loops. Pass statement is also employed for classes, functions, and empty control statements.

6. Lab Exercise:

a) Write a program to check whether the input year is a leap year or not.

The screenshot shows two windows from the Python IDLE 3.12.4 environment. The top window, titled 'exp2.py', contains a Python script to check if a given year is a leap year. The script prompts the user to enter a year and then checks if it is divisible by 400 or by 4 but not by 100. The bottom window, titled 'IDLE Shell 3.12.4', shows the execution of the script. It displays the Python version, the file path, and the output of the program, which is 'Entered year {} is a leap year' for the input '2024'.

```
exp2.py - C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py (3.12.4)
File Edit Format Run Options Window Help

print ("Check if given year is a leap year")
num=int(input("Entered year"))
if((num%400==0)or((num%4==0) and (num%100))):
    print("Entered year {} is a leap year")
else:
    print("Entered year {} is not a leap year")

IDLE Shell 3.12.4
File Edit Shell Debug Options Window Help

Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>>
==== RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py ====
Check if given year is a leap year
Entered year2024
Entered year {} is a leap year
>>>
```

Ln: 8 Col: 0

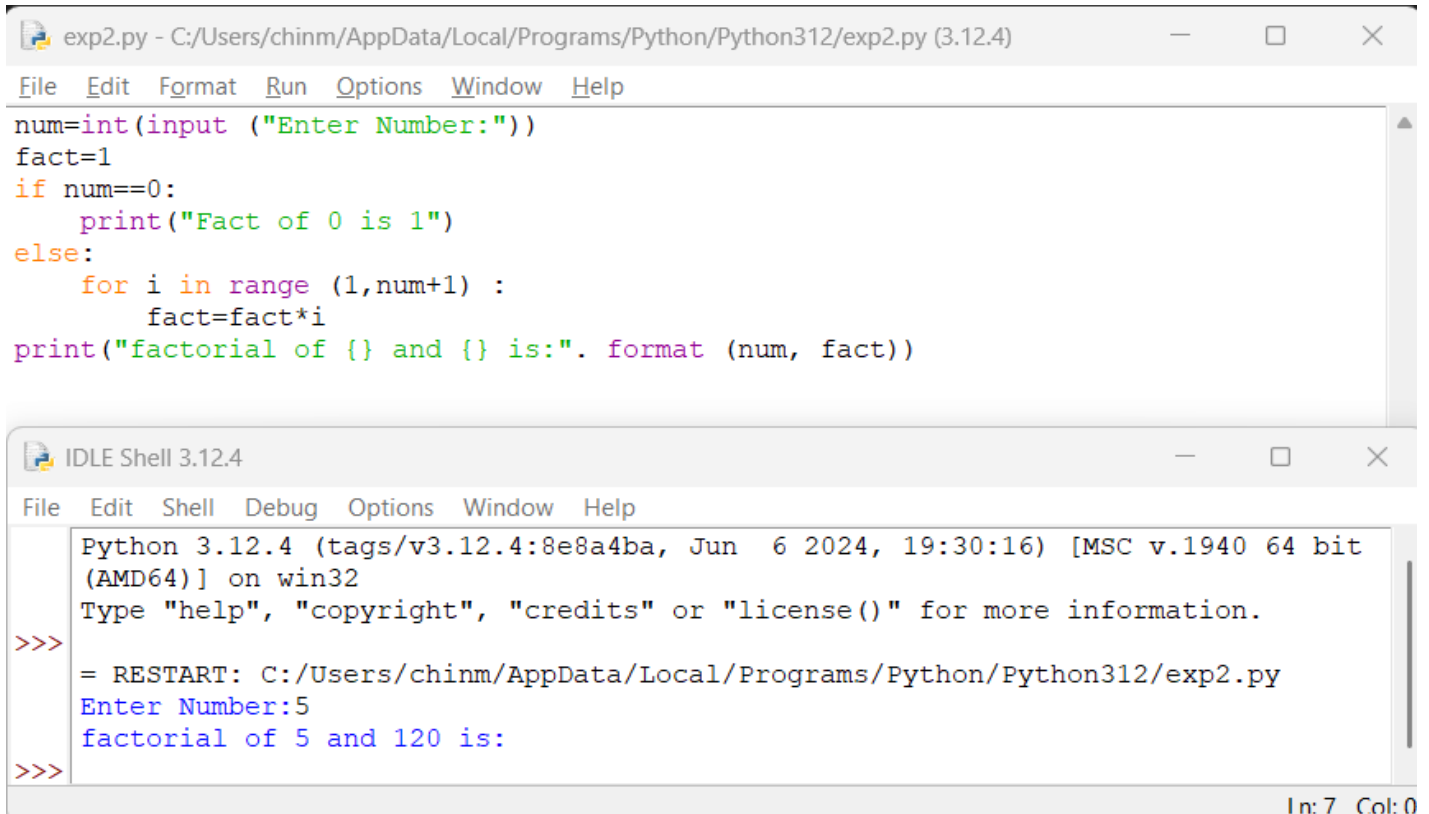
- b) Write a menu driven program for calculator for performing arithmetic operations / 1 to 7 weekdays display
(Note : use elif)

```
print("Perform Arithmetic calculation")
a = int(input("Enter 1st number: "))
b = int(input("Enter 2nd number: "))
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")
print("5. Modulus")
ch = int(input("Enter choice: "))

if ch == 1:
    c = a + b
    print("Addition of {} and {} is: {}".format(a, b, c))
elif ch == 2:
    c = a - b
    print("Subtraction of {} and {} is: {}".format(a, b, c))
elif ch == 3:
    c = a * b
    print("Multiplication of {} and {} is: {}".format(a, b, c))
elif ch == 4:
    if b != 0:
        c = a / b
        print("Division of {} and {} is: {}".format(a, b, c))
    else:
        print("Error! Division by zero is not possible.")
elif ch == 5:
    c = a % b
    print("Modulus of {} and {} is: {}".format(a, b, c))
else:
    print("Invalid choice")
```

```
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py
Perform Arithmetic calculation
Enter 1st number: 10
Enter 2nd number: 20
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter choice: 1
Addition of 10 and 20 is: 30
>>>
==== RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py ====
Perform Arithmetic calculation
Enter 1st number: 25
Enter 2nd number: 05
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter choice: 2
Subtraction of 25 and 5 is: 20
>>>
==== RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py ====
Perform Arithmetic calculation
Enter 1st number: 65
Enter 2nd number: 10
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter choice: 3
Multiplication of 65 and 10 is: 650
>>>
==== RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py ====
Perform Arithmetic calculation
Enter 1st number: 75
Enter 2nd number: 25
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter choice: 4
Division of 75 and 25 is: 3.0
>>>
==== RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py ====
Perform Arithmetic calculation
Enter 1st number: 83
Enter 2nd number: 8
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter choice: 5
Modulus of 83 and 8 is: 3
>>>
```

c)Write a program to check the factorial of a given number.



The screenshot shows two windows from the Python IDLE 3.12.4 environment. The top window, titled 'exp2.py - C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py (3.12.4)', contains the following Python code:

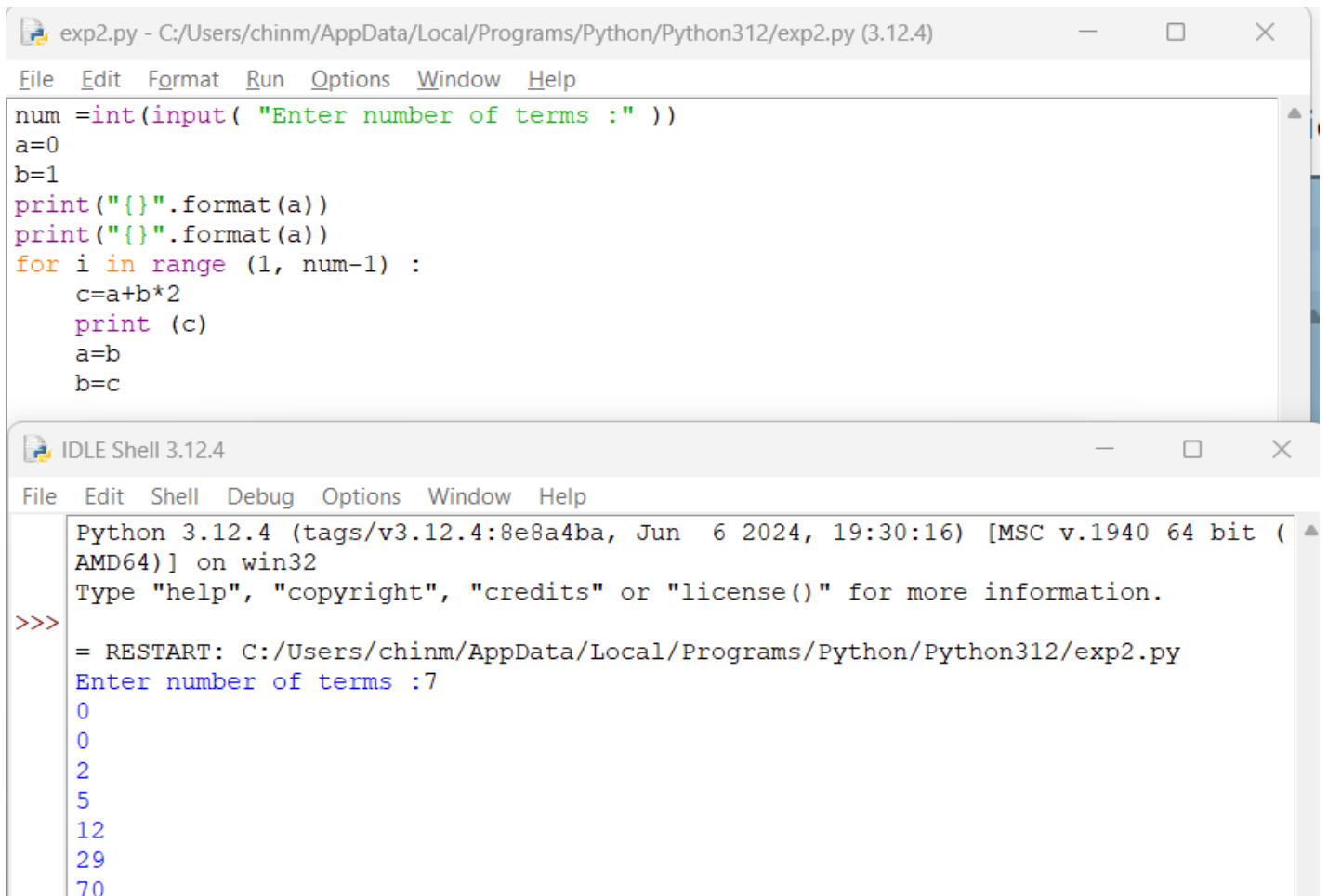
```
num=int(input ("Enter Number:"))
fact=1
if num==0:
    print("Fact of 0 is 1")
else:
    for i in range (1,num+1) :
        fact=fact*i
print("factorial of {} and {} is:". format (num, fact))
```

The bottom window, titled 'IDLE Shell 3.12.4', shows the execution of the program. It displays the Python version and system information, followed by the user's input and the program's output:

```
>>> Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py
Enter Number:5
factorial of 5 and 120 is:
>>>
```

The status bar at the bottom right of the shell window indicates 'In: 7 Col: 0'.

d)Write a program to print Fibonacci sequences/pell series for n numbers. Read the value of n from the user.



The screenshot shows two windows from the Python IDLE 3.12.4 environment. The top window, titled 'exp2.py - C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py (3.12.4)', contains the following Python code:

```
num =int(input( "Enter number of terms : " ))
a=0
b=1
print("{} ".format(a))
print("{} ".format(a))
for i in range (1, num-1) :
    c=a+b*2
    print (c)
    a=b
    b=c
```

The bottom window, titled 'IDLE Shell 3.12.4', shows the execution of the program. It displays the Python version and system information, followed by the user's input and the program's output:

```
>>> Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py
Enter number of terms :7
0
0
2
5
12
29
70
```

e) Write a program to find if the entered number by user is a palindrome number/Armstrong number/gcd-lcm/prime number/perfect square etc. (any)

The screenshot shows a Python IDE window titled 'exp2.py - C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py (3.12.4)'. The code defines a function to check if a number is a palindrome by reversing its digits. Below the code, the 'IDLE Shell 3.12.4' window shows the program's execution. It prompts the user to enter a number. The first input is '1221', and the program outputs 'The number is a palindrome'. The second input is '1234', and the program outputs 'The number is not palindrome'.

```

exp2.py - C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py (3.12.4)
File Edit Format Run Options Window Help
n=int(input("Enter Number: "))
temp=n
rev=0
while(n>0):
    dig=n%10
    rev=rev*10+dig
    n=n//10
if(temp==rev):
    print("The number is a palindrome")
else:
    print("The number is not palindrome")

IDLE Shell 3.12.4
File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py
Enter Number: 1221
The number is a palindrome
>>>
==== RESTART: C:/Users/chinm/AppData/Local/Programs/Python/Python312/exp2.py ====
Enter Number: 1234
The number is not palindrome

```

7. Post Experiment exercise:

7.1 Extended Theory

a) Write python programming to demonstrate use of break, continue and pass statement

The screenshot shows two windows in a Python IDE. The left window, titled 'IDLE Shell 3.13.1', shows the execution of a program that prints the word 'python' vertically. The right window, titled 'exp2PE_7a.py - C:/Users/MAP/Documents/b47/exp2PE_7a.py (...)', shows the source code. The code uses a 'for' loop to iterate over the string 'python'. It uses 'continue' to skip the rest of the loop body when the character is 'h', and 'pass' to skip the rest of the loop body when the character is 'n'. The loop ends with a 'print("end")' statement.

```

IDLE Shell 3.13.1
File Edit Shell Debug Options Window Help
Python 3.13.1 (tags/v3.13.1:0671451, Dec 3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/MAP/Documents/b47/exp2PE_7a.py =====
p
y
t
o
n
end
>>>

exp2PE_7a.py - C:/Users/MAP/Documents/b47/exp2PE_7a.py (...)
File Edit Format Run Options Window Help
s="python"
i=0
for i in range(0,len(s)):
    if(s[i]=='h'):
        continue
    print("\t{}".format(s[i]))
pass
print("end")

```

b) write a program to calculate electricity bill (accept number of units from the user) according to following criteria

Unit price

for 100 units no charge

for next 100 units Rs. 5 per unit

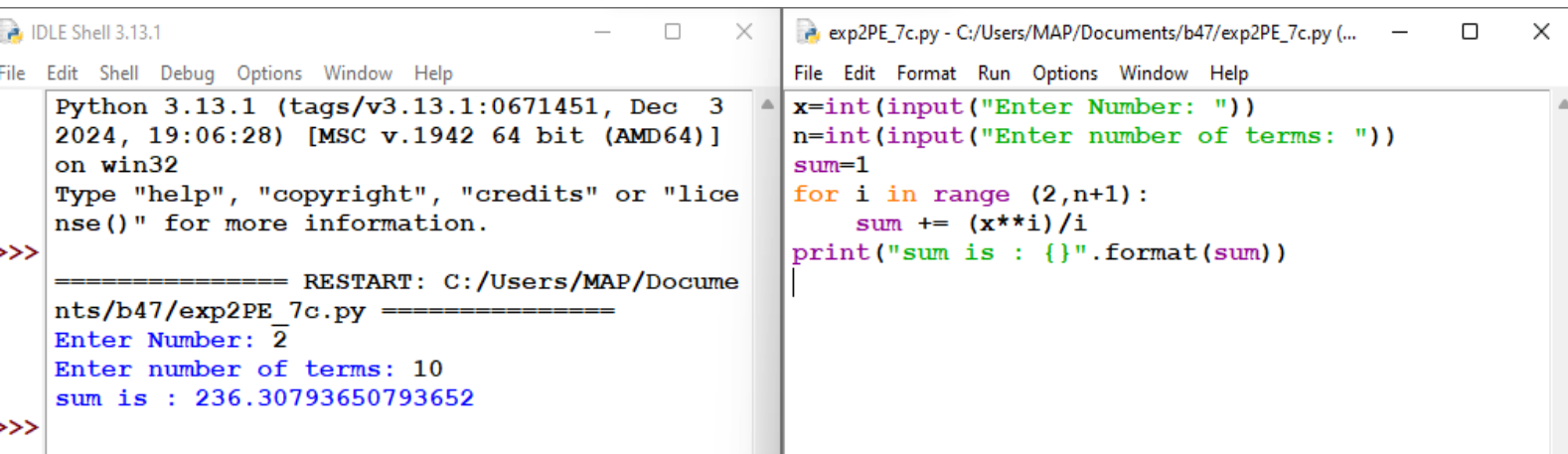
after 200 units Rs. 10 per unit

for example input is 350 units ,, bill will be 2000

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec 3 2024, 19:06:28) [MSC v.1942 64 bit (AMD 64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> ===== RESTART: C:/Users/MAP/Documents/b47/exp2PE_7b.py =====
Enter number of units: 350
Electricity charge is 2000
>>> = RESTART: C:/Users/MAP/Documents/b47/exp2PE_7b.py
Enter number of units: 150
Electricity charge is 250
>>> = RESTART: C:/Users/MAP/Documents/b47/exp2PE_7b.py
Enter number of units: 50
No Charge
>>> = RESTART: C:/Users/MAP/Documents/b47/exp2PE_7b.py
Enter number of units: -1
Invalid input
>>> |
```

```
exp2PE_7b.py - C:/Users/MAP/Documents/b47/exp2PE_7b.py (3.13.1)
File Edit Format Run Options Window Help
u=int(input("Enter number of units: "))
if(u<0):
    print("Invalid input")
else:
    if(u<=100):
        print("No Charge")
    elif(u>100 and u<=200):
        c = (u-100)*5
        print("Electricity charge is {}".format(c))
    else:
        a=5*100
        b=10*(u-200)
        c=a+b
        print("Electricity charge is {}".format(c))
```


c) Write a program to calculate the summation for the given series $1 + x^2/2 + x^3/3 + x^n/n$



The image shows two windows from the Python IDLE environment. The left window is the 'IDLE Shell 3.13.1' showing the program's execution. The right window is the 'exp2PE_7c.py' editor showing the source code.

IDLE Shell 3.13.1 Output:

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec 3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/MAP/Documents/b47/exp2PE_7c.py =====
Enter Number: 2
Enter number of terms: 10
sum is : 236.30793650793652
>>>
```

exp2PE_7c.py Source Code:

```
x=int(input("Enter Number: "))
n=int(input("Enter number of terms: "))
sum=1
for i in range (2,n+1):
    sum += (x**i)/i
print("sum is : {}".format(sum))
```

7.2 Results/ observations / program output:

Present the program input/output results and comment on the same

7.3 Conclusion :

1. Write what was performed in the program (s).
2. What is the significance of program (s)?

8. References :

8.2.1 <https://www.javatpoint.com/python-applications>

8.2.2 <https://www.geeksforgeeks.org/python-output-using-print-function/?ref=lbp>

